

HDOC - Day 9

Menu-Driven C Programming Activities

Prepare the flowchart, pseudocode, algorithm, test case table, C code, compilation and execution, and test each program against the prepared test cases.

Question 1: Swapping Two Numbers

Write a menu-driven C program to input two numbers and swap them using a third variable and without using a third variable. Display the values before and after swapping.

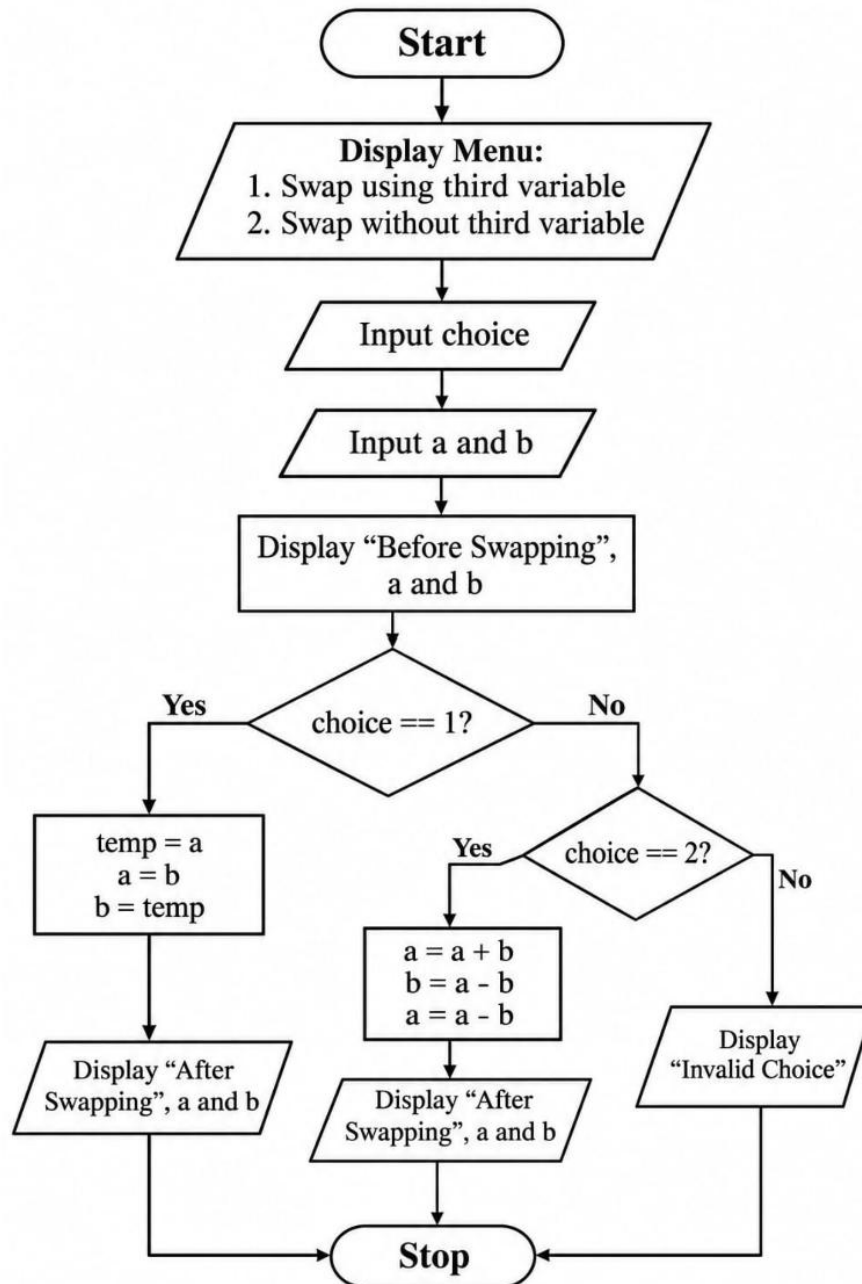
Algorithm

1. Start.
2. Display menu: 1. Swap using third variable 2. Swap without third variable.
3. Input choice.
4. Input two integers a and b.
5. Display a and b before swapping.
6. If choice = 1, use temp = a, a = b, b = temp.
7. Else if choice = 2, use a = a + b, b = a - b, a = a - b.
8. Otherwise display Invalid choice.
9. Display a and b after swapping.
10. Stop.

Pseudocode

```
BEGIN
  DISPLAY menu
  INPUT choice
  INPUT a, b
  DISPLAY a, b before swapping
  IF choice = 1 THEN
    temp <- a
    a <- b
    b <- temp
  ELSE IF choice = 2 THEN
    a <- a + b
    b <- a - b
    a <- a - b
  ELSE
    DISPLAY "Invalid choice"
  END IF
  DISPLAY a, b after swapping
END
```

Flowchart



Test Case Table

Test	a	b	Choice	Expected output
1	10	20	1	Before: 10,20; After: 20,10
2	5	8	2	Before: 5,8; After: 8,5
3	-4	7	1	Before: -4,7; After: 7,-4
4	3	9	3	Invalid choice

C Program

```
#include <stdio.h>

int main()
{
    int a, b, temp, choice;

    printf("1. Swap using third variable\n");
    printf("2. Swap without third variable\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    printf("Before swapping: a = %d, b = %d\n", a, b);

    if (choice == 1)
    {
        temp = a;
        a = b;
        b = temp;
    }
    else if (choice == 2)
    {
        a = a + b;
        b = a - b;
        a = a - b;
    }
    else
    {
        printf("Invalid choice\n");
        return 0;
    }

    printf("After swapping: a = %d, b = %d\n", a, b);
    return 0;
}
```

Compilation and Execution

```
(kali@kali)-[~/hdoc9.c]
$ gcc swap.c -o swap
```

```
(kali@kali)-[~/hdoc9.c]
$ ./swap
MENU
1. Swap using third variable
2. Swap without using third variable
Enter your choice: 1
Enter two numbers: 10 20

Before Swapping:
a = 10, b = 20

After Swapping using third variable:
a = 20, b = 10

(kali@kali)-[~/hdoc9.c]
$ ./swap
MENU
1. Swap using third variable
2. Swap without using third variable
Enter your choice: 2
Enter two numbers: 25 15

Before Swapping:
a = 25, b = 15

After Swapping without third variable:
a = 15, b = 25
```

Question 2: Sum of Natural Numbers and Their Squares

Write a menu-driven C program to find the sum of the first n natural numbers and the sum of squares of the first n natural numbers.

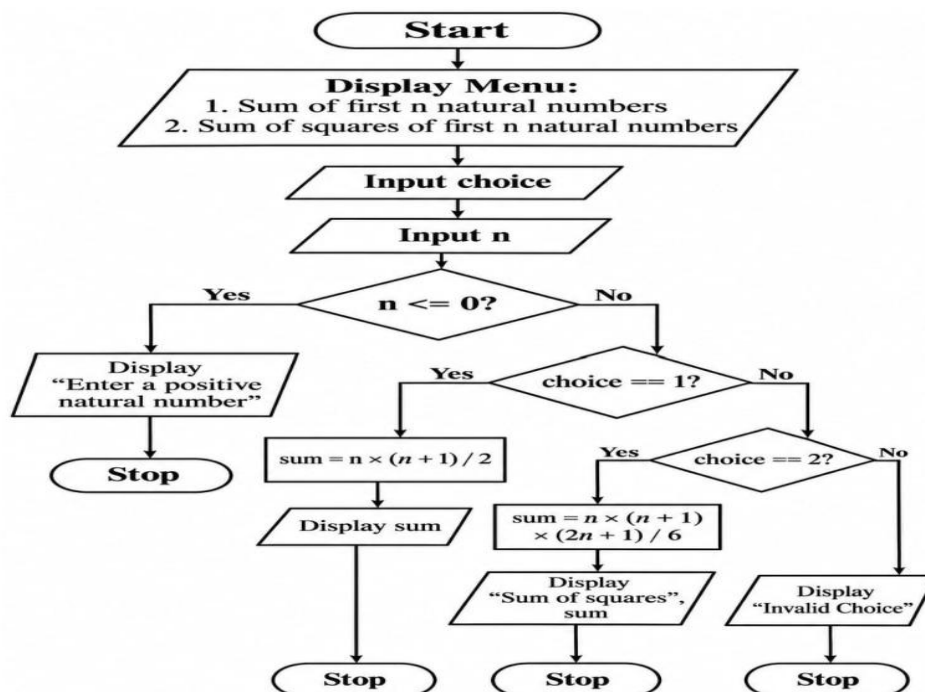
Algorithm

1. Start.
2. Display menu: 1. Sum of first n natural numbers 2. Sum of squares.
3. Input choice and n.
4. If $n \leq 0$, display a message that n must be positive.
5. If choice = 1, calculate $\text{sum} = n(n + 1)/2$.
6. Else if choice = 2, calculate $\text{sum} = n(n + 1)(2n + 1)/6$.
7. Otherwise display Invalid choice.
8. Display the result.
9. Stop.

Pseudocode

```
BEGIN
  DISPLAY menu
  INPUT choice
  INPUT n
  IF n <= 0 THEN
    DISPLAY "Enter a positive natural number"
  ELSE IF choice = 1 THEN
    sum <- n * (n + 1) / 2
    DISPLAY sum
  ELSE IF choice = 2 THEN
    sum <- n * (n + 1) * (2*n + 1) / 6
    DISPLAY sum
  ELSE
    DISPLAY "Invalid choice"
  END IF
END
```

Flowchart



Test Case Table

Test	n	Choice	Expected output
1	5	1	Sum = 15
2	5	2	Sum of squares = 55
3	1	1	Sum = 1
4	0	1	Enter a positive natural number
5	5	3	Invalid choice

C Program

```
#include <stdio.h>

int main()
{
    int n, choice, sum;

    printf("1. Sum of first n natural numbers\n");
    printf("2. Sum of squares of first n natural numbers\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("Enter n: ");
    scanf("%d", &n);

    if (n <= 0)
        printf("Enter a positive natural number\n");
    else if (choice == 1)
    {
        sum = n * (n + 1) / 2;
        printf("Sum = %d\n", sum);
    }
    else if (choice == 2)
    {
        sum = n * (n + 1) * (2 * n + 1) / 6;
        printf("Sum of squares = %d\n", sum);
    }
    else
        printf("Invalid choice\n");

    return 0;
}
```

Compilation and Execution

```
(kali@kali)-[~/hdoc9.c]
$ gcc sum.c -o SUM
```

```
(kali@kali)-[~/hdoc9.c]
$ ./sum
MENU
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
Enter your choice: 2
Enter n: 5
Sum of squares = 55

(kali@kali)-[~/hdoc9.c]
$ ./sum
MENU
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
Enter your choice: 1
Enter n: 5
Sum = 15
```

Question 3: Interest and Payable Amount

Write a menu-driven C program to calculate simple interest, compound amount, compound interest, and total payable amount for a given principal, rate, and time.

Formulae Used

- Simple Interest (SI) = $(P \times R \times T) / 100$
- Compound Amount (A) = $P \times (1 + R/100)^T$
- Compound Interest (CI) = $A - P$
- Total Payable Amount = $P + SI$ (for the simple-interest case)

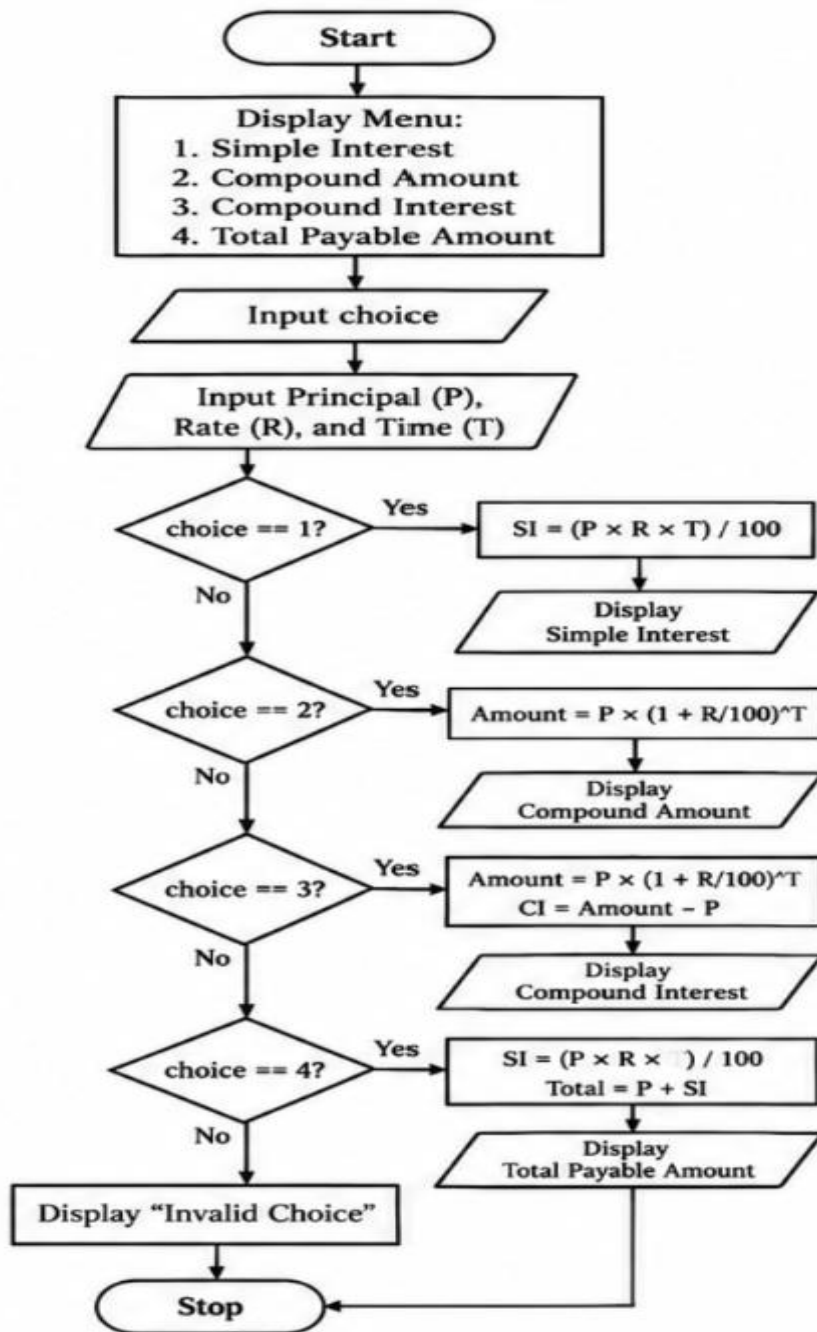
Algorithm

1. Start.
2. Display menu with four choices: Simple Interest, Compound Amount, Compound Interest, Total Payable Amount.
3. Input choice, principal P, rate R, and time T.
4. If choice = 1, calculate $SI = (P \times R \times T) / 100$.
5. If choice = 2, calculate compound amount $A = P \times (1 + R/100)^T$.
6. If choice = 3, calculate A and then $CI = A - P$.
7. If choice = 4, calculate SI and total = $P + SI$.
8. Otherwise display Invalid choice.
9. Display the calculated result.
10. Stop.

Pseudocode

```
BEGIN
  DISPLAY menu
  INPUT choice, P, R, T
  IF choice = 1 THEN
    SI <- (P * R * T) / 100
    DISPLAY SI
  ELSE IF choice = 2 THEN
    A <- P * (1 + R/100)^T
    DISPLAY A
  ELSE IF choice = 3 THEN
    A <- P * (1 + R/100)^T
    CI <- A - P
    DISPLAY CI
  ELSE IF choice = 4 THEN
    SI <- (P * R * T) / 100
    total <- P + SI
    DISPLAY total
  ELSE
    DISPLAY "Invalid choice"
  END IF
END
```

Flowchart



Test Case Table

Test	P	R	T	Choice	Expected output
1	1000	10	2	1	SI = 200.00
2	1000	10	2	2	Amount = 1210.00
3	1000	10	2	3	CI = 210.00
4	1000	10	2	4	Total = 1200.00
5	1000	10	2	5	Invalid choice

C Program

```
#include <stdio.h>
#include <math.h>

int main()
{
    int choice;
    float p, r, t, si, amount, ci, total;

    printf("1. Simple Interest\n");
    printf("2. Compound Amount\n");
    printf("3. Compound Interest\n");
    printf("4. Total Payable Amount\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("Enter principal amount: ");
    scanf("%f", &p);

    printf("Enter rate of interest: ");
    scanf("%f", &r);

    printf("Enter time: ");
    scanf("%f", &t);

    if (choice == 1)
    {
        si = (p * r * t) / 100;
        printf("Simple Interest = %.2f\n", si);
    }
    else if (choice == 2)
    {
        amount = p * pow(1 + r / 100, t);
        printf("Compound Amount = %.2f\n", amount);
    }
    else if (choice == 3)
    {
        amount = p * pow(1 + r / 100, t);
        ci = amount - p;
        printf("Compound Interest = %.2f\n", ci);
    }
    else if (choice == 4)
    {
        si = (p * r * t) / 100;
        total = p + si;
        printf("Total Payable Amount = %.2f\n", total);
    }
    else
        printf("Invalid choice\n");

    return 0;
}
```


Compilation and Execution

```
(kali㉿kali)-[~/hdoc9.c]  
$ gcc intrest.c -o intrest -lm
```

```
(kali㉿kali)-[~/hdoc9.c]  
$ ./intrest  
MENU  
1. Simple Interest  
2. Compound Amount  
3. Compound Interest  
4. Total Payable Amount  
Enter your choice: 1  
Enter principal amount: 10000  
Enter rate of interest: 5  
Enter time: 2  
Simple Interest = 1000.00
```

```
(kali㉿kali)-[~/hdoc9.c]  
$ ./intrest  
MENU  
1. Simple Interest  
2. Compound Amount  
3. Compound Interest  
4. Total Payable Amount  
Enter your choice: 2  
Enter principal amount: 10000  
Enter rate of interest: 5  
Enter time: 2  
Compound Amount = 11025.00
```

```
(kali㉿kali)-[~/hdoc9.c]  
$ ./intrest  
MENU  
1. Simple Interest  
2. Compound Amount  
3. Compound Interest  
4. Total Payable Amount  
Enter your choice: 4  
Enter principal amount: 10000  
Enter rate of interest: 5  
Enter time: 2  
Total Payable Amount = 11000.00
```

```
(kali㉿kali)-[~/hdoc9.c]  
$ ./intrest  
MENU  
1. Simple Interest  
2. Compound Amount  
3. Compound Interest  
4. Total Payable Amount  
Enter your choice: 3  
Enter principal amount: 10000  
Enter rate of interest: 5  
Enter time: 2  
Compound Interest = 1025.00
```